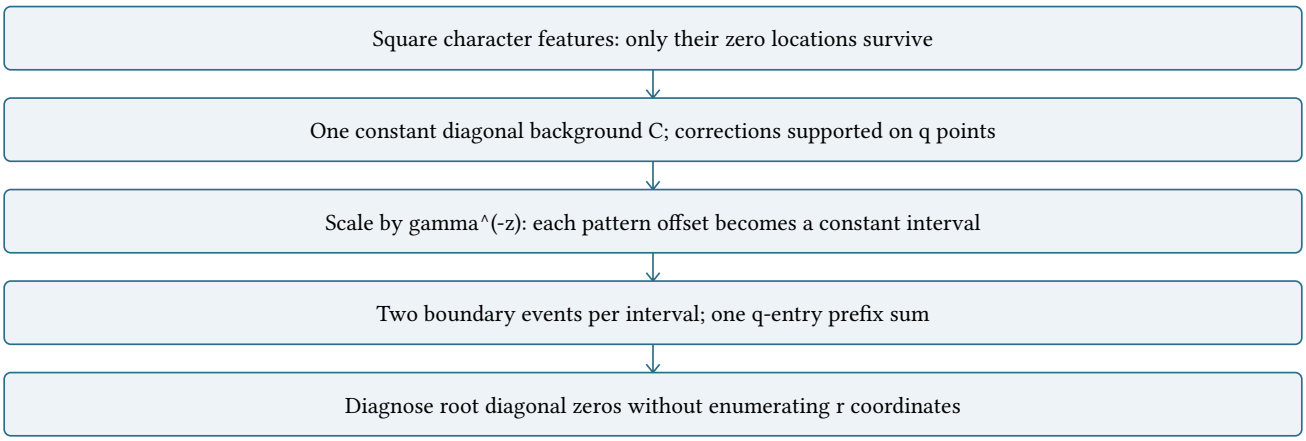
Compact diagonal of the public feature Gram

2026-11-02 · Derived identities / static checks · Numerical trials: zero

Actual source. The original compiler uses weighted character products with patterns P of size 1, 2 or 3, shifts 0 through $\ell-1$, where $\ell=q-\max(P)$. Because $\chi(x)^2$ is 1 except at zero, the squared feature only records its zeros. Distinct offsets ensure at most one zero factor per shifted feature.



Representation. Let $H=\Sigma_p\, w_p(\gamma^\ell-1)/(\gamma-1)$, $C=w_0+H$, and $z=-k \bmod r$. Then $B_{kk}=C-\Delta_z$ for z in $0,\dots,q-1$, and $B_{kk}=C$ otherwise. Each Δ_z sums $w_p\,\gamma^{(z-d)}$ over offsets d whose interval $[d,d+\ell)$ contains z. Scaling by $\gamma^{(-z)}$ makes each interval constant, so endpoint events recover all corrections.

Derived build cost	Count
Pattern count M	$1+q(q-1)/2$
Offset incidences L	$1+2(q-1)+3(q-1)(q-2)/2$
Field products	$3q-1+2M+L$
Field additions/subtractions	$2+2M+2L+q$
Field inversions	2
Arithmetic workspace	O(q), excluding input weights / meter history

Root consequence. For $A=D^{-1}JBJ$, exceptional diagonal indices are $r-1$ and $0,\dots,q-2$. If C is nonzero, all bulk diagonals are nonzero, so a root zero-column prefix has length at most $q-1$. A zero diagonal is only a necessary condition for a zero column. These statements do not certify nonsingularity.

Weight diagnostic. With other weights fixed, bad nonzero constant weights are precisely the nonzero values among $-H$ and Δ_z-H. There are at most $q+1$. If q is at most $r-3$, some allowed nonzero weight avoids every diagonal zero. The code returns this set only; it does not modify or select frozen weights.

Boundary. Schur diagonals subtract $B_i^I\,E^{-1}\,B^I_i$. Closure of the compact root representation under this update is not established. The full matrix, its inverse and extractor have not been compressed by this diagonal identity.

Checks / next. Five Python sources parse; the original constructor copy and pattern-expression AST match; static inspection finds no field-size sweep, character exponentiation or source call in the builder. The independent direct character checker is implemented but unexecuted. Numerical/corruption/heap/ performance trials remain zero. Next bind a root-only pivot proposal to the constructor and retain fallback; investigate Schur diagonal updates separately.

Sources. DERIVATION.md, diagonal.py, independent_check.py, reference_constructor.py; NIST DLMF 27.9 for character values. Original public feature definition: direct-cyclic-companion/DERIVATION.md.